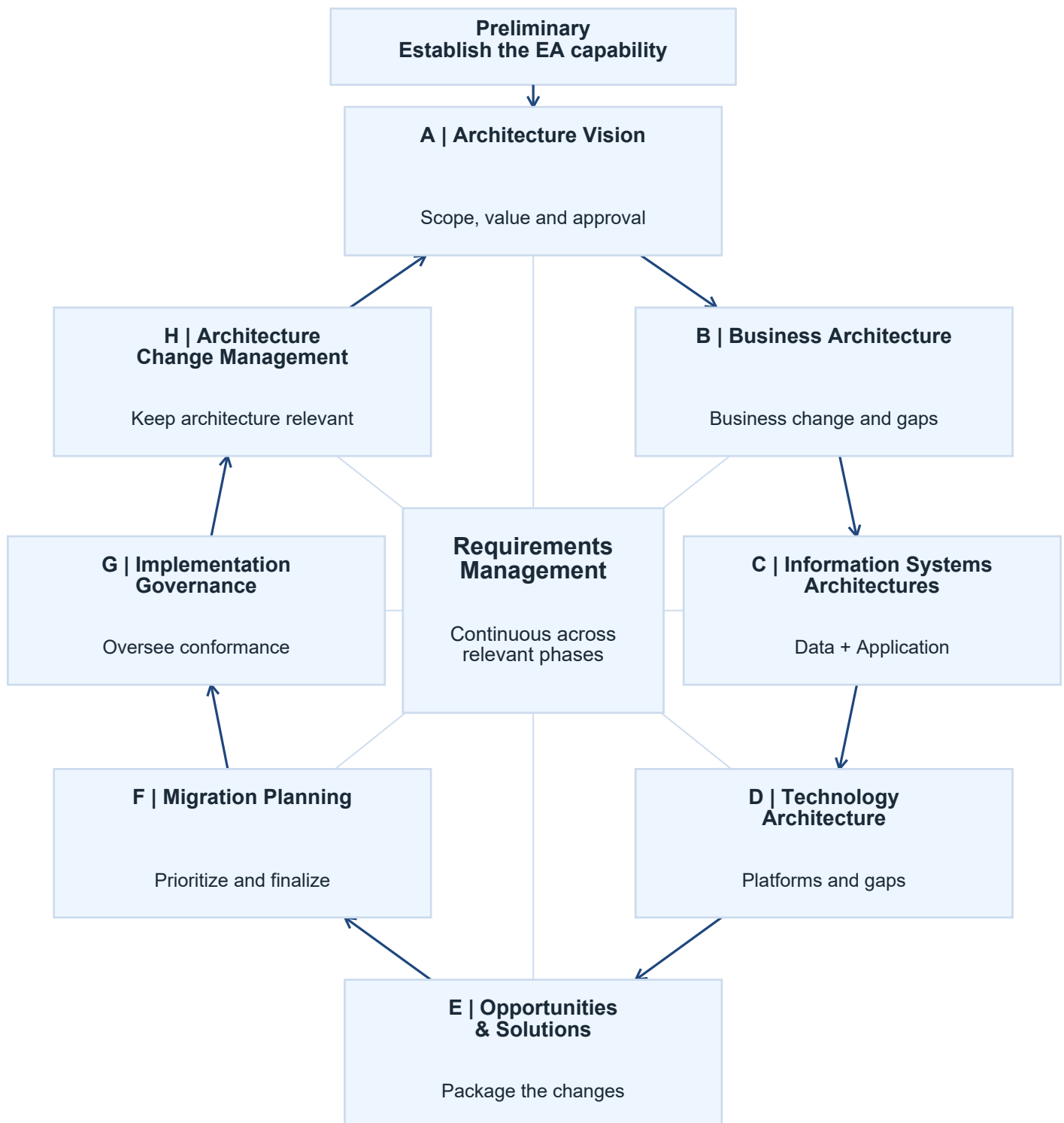


Architecture Development Method



REMEMBER

The cycle is iterative, not a mandatory waterfall. Revisit work within phases, between phases and across cycles. Dashed lines show continuous Requirements Management. C covers both Data and Application Architecture.

Inputs and outputs at a glance

PHASE	KEY INPUT	KEY OUTPUT
Preliminary Preliminary Phase	Business/IT strategies, board plans, business principles, goals and drivers; existing Architecture Capability.	Organizational Model for Enterprise Architecture: scope, maturity gaps, roles, constraints, budget and governance.
A Architecture Vision	Request for Architecture Work; business principles, goals and drivers.	Approved Statement of Architecture Work: scope, vision, plan and schedule.
B Business Architecture	Business goals, drivers, principles; Capability Assessment and Communications Plan.	Draft ADD: approved Target Business Architecture; approved Baseline, if appropriate; stakeholder views.
C: Data Data Architecture	Business Architecture gaps and technical requirements in the draft Requirements Specification.	Draft ADD: approved Target Data Architecture; approved Baseline, if appropriate; stakeholder views.
C: Application Application Architecture	Business and, if available, Data Architecture gaps and relevant technical requirements.	Draft ADD: approved Target Application Architecture; approved Baseline, if appropriate; stakeholder views.
D Technology Architecture	Business, Data and Application gaps, requirements and candidate roadmap components.	Draft ADD: approved Target Technology Architecture; approved Baseline, if appropriate; stakeholder views.
E Opportunities & Solutions	B/C/D draft ADD, requirements, gap analyses and candidate roadmap components.	Architecture Roadmap: work packages, dependencies, value, transitions and implementation recommendations (including SBBs).
F Migration Planning	Draft Architecture Roadmap: work packages, transitions and Implementation Factor catalog.	Approved Implementation and Migration Plan: strategy, project/portfolio breakdown, milestones, resources and costs.
G Implementation Governance	ADD and Architecture Requirements Specification; Architecture Roadmap.	Signed Architecture Contract; Compliance Assessments; Change Requests.
H Architecture Change Management	Current ADD, Requirements Specification and Architecture Roadmap.	Architecture updates for maintenance changes.
Requirements Architecture Requirements Management	Populated Architecture Repository; Organizational Model for EA; tailored framework.	Requirements Impact Assessment.

REMEMBER

ADD = Architecture Definition Document; ABB = Architecture Building Block. Deliverables are reviewed work products; artifacts describe architecture through catalogs, matrices and diagrams. Select these to address stakeholder concerns; outputs are tailored, not a universal checklist. C covers Data and Application Architecture.

Preliminary Phase

OBJECTIVES

Determine the desired Architecture Capability and establish it: organization, governance, processes, tools and Architecture Principles.

APPROACH

Define the enterprise, drivers, organizational context and work requirements. Set principles and framework relationships; assess maturity and target capability. Revisit when an engagement needs a different capability.

INPUTS

- Business/IT strategies, board plans, business principles, goals and drivers; existing Architecture Capability.
- Business management, governance and legal frameworks; partnership and contract agreements.
- Existing Organizational Model for EA, Architecture Framework and Architecture Repository, if any.
- Architecture guidance and other frameworks, where required.

MAIN STEPS

- 1 Scope the enterprise organizations impacted.
- 2 Confirm governance and support frameworks.
- 3 Define and establish Enterprise Architecture team and organization.
- 4 Identify and establish Architecture Principles.
- 5 Tailor the TOGAF framework and other selected architecture frameworks, if any.
- 6 Develop a strategy and implementation plan for tools and techniques.

OUTPUTS / DELIVERABLES

- Organizational Model for Enterprise Architecture: scope, maturity gaps, roles, constraints, budget and governance.
- Tailored Architecture Framework: method, content, Architecture Principles and configured tools.
- Initial Architecture Repository.
- Restated business principles, goals and drivers.
- Request for Architecture Work (optional).
- Architecture Governance Framework.
- Architecture of the EA Capability.

ARTIFACTS / WORK PRODUCTS

- Principles catalog: principle name, statement, rationale and implications. The Organizational Model and Tailored Architecture Framework are deliverables, not catalog types.

REMEMBER

Establish the capability to do architecture. Scope, governance, principles, terminology, process and content tailoring belong here.

Architecture Vision

OBJECTIVES

Develop a high-level vision of capabilities and business value. Obtain approval for the Statement of Architecture Work to develop and deploy that architecture.

APPROACH

Engage stakeholders early. Establish scope, concerns, constraints, value and readiness before detailed design. Use business scenarios where helpful; agree sponsorship, communications and approval.

INPUTS

- Request for Architecture Work; business principles, goals and drivers.
- Organizational Model for EA; Tailored Architecture Framework and principles.
- Populated Architecture Repository: existing descriptions, baselines and ABBs.
- Relevant architecture reference materials.

OUTPUTS / DELIVERABLES

- Approved Statement of Architecture Work: scope, vision, plan and schedule.
- Refined business principles, goals and drivers.
- Architecture Principles.
- Capability Assessment.
- Tailored Architecture Framework for the engagement.
- Architecture Vision: problem, summary views and high-level stakeholder requirements.
- Draft Architecture Definition Document (ADD).
- Communications Plan.
- Architecture Repository additions.

MAIN STEPS

- 1 Establish the architecture project.
- 2 Identify stakeholders, concerns, and business requirements.
- 3 Confirm and elaborate business goals, business drivers, and constraints.
- 4 Evaluate capabilities.
- 5 Assess readiness for business transformation.
- 6 Define scope.
- 7 Confirm and elaborate Architecture Principles, including business principles.
- 8 Develop Architecture Vision.
- 9 Define the Target Architecture value propositions and KPIs.
- 10 Identify business transformation risks and mitigation activities.
- 11 Develop Statement of Architecture Work; secure approval.

ARTIFACTS / WORK PRODUCTS

- Stakeholder catalog; Value Chain diagram; Solution Concept diagram; Business Model diagram; Business Capability Map; Value Stream Map. Select views for actual stakeholder concerns.

REMEMBER

Request for Architecture Work starts the work; the approved Statement of Architecture Work defines the agreed engagement. These are different documents.

Business Architecture

OBJECTIVES

Develop the Target Business Architecture that addresses business goals, the Architecture Vision and stakeholder concerns. Identify candidate roadmap components from gaps.

APPROACH

Use capabilities, value streams, organization, processes and information to explain business change. Choose viewpoints for stakeholder concerns and reuse suitable models; model only to the depth needed.

INPUTS

- Business goals, drivers, principles; Capability Assessment and Communications Plan.
- Architecture Vision; approved Statement of Architecture Work; draft ADD.
- Organizational Model for EA; Tailored Architecture Framework; domain principles.
- Repository building blocks, reference models and standards.
- Request for Architecture Work; Capability Assessment; Communications Plan; external architecture guidance. Enterprise Continuum.

OUTPUTS / DELIVERABLES

- Draft ADD: approved Target Business Architecture; approved Baseline, if appropriate; stakeholder views.
- Draft Architecture Requirements Specification: gaps and domain requirements, including cross-domain implications.
- Business content: organization, goals, functions, capabilities, services, products, processes and roles.
- Business Architecture components of the Architecture Roadmap.
- Refined Phase A deliverables and domain principles, where applicable.

MAIN STEPS

- 1 Select reference models, viewpoints, and tools.
- 2 Develop Baseline Business Architecture Description.
- 3 Develop Target Business Architecture Description.
- 4 Perform gap analysis.
- 5 Define candidate roadmap components.
- 6 Resolve impacts across the Architecture Landscape.
- 7 Conduct formal stakeholder review.
- 8 Finalize the Business Architecture.
- 9 Create/update the Architecture Definition Document.

ARTIFACTS / WORK PRODUCTS

- Catalogs: Organization/Actor, Driver/Goal/Objective, Role, Business Service/Function. Matrices: Actor/Role, Business Interaction, Value Stream/Capability. Diagrams: Business Footprint, Process Flow; Business Capability Map.

REMEMBER

Capabilities, value streams, organization, processes and information are business concerns. Models support stakeholder decisions.

Data Architecture

OBJECTIVES

Develop the Target Data Architecture that enables the Business Architecture and Architecture Vision. Identify candidate roadmap components from gaps.

APPROACH

Cover data at rest, in motion, in use and open data. Address ownership, management, migration and governance. Iterate Data and Application Architecture in the order suited to the engagement.

INPUTS

- Business Architecture gaps and technical requirements in the draft Requirements Specification.
- Architecture Vision; Statement of Architecture Work; draft ADD; Business roadmap components.
- Organizational Model for EA; Tailored Architecture Framework; domain principles.
- Repository building blocks, reference models and standards.
- Request for Architecture Work; Capability Assessment; Communications Plan; external architecture guidance.

MAIN STEPS

- 1 Select reference models, viewpoints, and tools.
- 2 Develop Baseline Data Architecture Description.
- 3 Develop Target Data Architecture Description.
- 4 Perform gap analysis.
- 5 Define candidate roadmap components.
- 6 Resolve impacts across the Architecture Landscape.
- 7 Conduct formal stakeholder review.
- 8 Finalize the Data Architecture.
- 9 Create/update the Architecture Definition Document.

OUTPUTS / DELIVERABLES

- Draft ADD: approved Target Data Architecture; approved Baseline, if appropriate; stakeholder views.
- Draft Architecture Requirements Specification: gaps and domain requirements, including cross-domain implications.
- Data content: business/logical data models, data management; interoperability, technology constraints and updated business/application requirements.
- Data Architecture components of the Architecture Roadmap.
- Refined Phase A deliverables and domain principles, where applicable.

ARTIFACTS / WORK PRODUCTS

- Data Entity/Data Component catalog; Data Entity/Business Function and Application/Data matrices; Conceptual Data, Logical Data, Data Dissemination, Data Security, Data Migration and Data Lifecycle diagrams.

REMEMBER

Model the data and its use, sharing and management. Data and Application Architecture may iterate with each other.

Application Architecture

OBJECTIVES

Develop the Target Application Architecture that enables the Business Architecture and Architecture Vision. Identify candidate roadmap components from gaps.

APPROACH

Describe applications, services, interactions and business support, not detailed software design. Reuse application models; use Data Architecture inputs if available and resolve cross-domain dependencies.

INPUTS

- Business and, if available, Data Architecture gaps and relevant technical requirements.
- Architecture Vision; Statement of Architecture Work; draft ADD; Business/Data roadmap components, if available.
- Organizational Model for EA; Tailored Architecture Framework; domain principles.
- Repository building blocks, reference models and standards.
- Request for Architecture Work; Capability Assessment; Communications Plan; external architecture guidance.

MAIN STEPS

- 1 Select reference models, viewpoints, and tools.
- 2 Develop Baseline Application Architecture Description.
- 3 Develop Target Application Architecture Description.
- 4 Perform gap analysis.
- 5 Define candidate roadmap components.
- 6 Resolve impacts across the Architecture Landscape.
- 7 Conduct formal stakeholder review.
- 8 Finalize the Application Architecture.
- 9 Create/update the Architecture Definition Document.

OUTPUTS / DELIVERABLES

- Draft ADD: approved Target Application Architecture; approved Baseline, if appropriate; stakeholder views.
- Draft Architecture Requirements Specification: gaps and domain requirements, including cross-domain implications.
- Application content: portfolio and interfaces; interoperability, technology constraints and updated business/data requirements.
- Application Architecture components of the Architecture Roadmap.
- Refined Phase A deliverables and domain principles, where applicable.

ARTIFACTS / WORK PRODUCTS

- Application Portfolio and Interface catalogs; Application/Organization, Role/Application and Application/Function matrices; Application Communication, Application Use-Case and Application Migration diagrams.

REMEMBER

Describe application components and their interactions. Data Architecture inputs are used if available; Phase C is not a rigid data-first sequence.

Technology Architecture

OBJECTIVES

Develop the Target Technology Architecture that enables the Business, Data and Application Architectures. Identify candidate roadmap components from gaps.

APPROACH

Describe technology services, platforms, environments, locations, load and networks. Evaluate emerging technology opportunities against business needs, standards, security and interoperability.

INPUTS

- Business, Data and Application gaps, requirements and candidate roadmap components.
- Architecture Vision; Statement of Architecture Work; draft ADD and Requirements Specification.
- Organizational Model for EA; Tailored Architecture Framework; domain principles.
- Repository building blocks, reference models and standards. Candidate product information.
- Request for Architecture Work; Capability Assessment; Communications Plan; external architecture guidance.

MAIN STEPS

- 1 Select reference models, viewpoints, and tools.
- 2 Develop Baseline Technology Architecture Description.
- 3 Develop Target Technology Architecture Description.
- 4 Perform gap analysis.
- 5 Define candidate roadmap components.
- 6 Resolve impacts across the Architecture Landscape.
- 7 Conduct formal stakeholder review.
- 8 Finalize the Technology Architecture.
- 9 Create/update the Architecture Definition Document.

OUTPUTS / DELIVERABLES

- Draft ADD: approved Target Technology Architecture; approved Baseline, if appropriate; stakeholder views.
- Draft Architecture Requirements Specification: gaps and domain requirements, including cross-domain implications.
- Technology content: platforms, environments, load, networks; updated technology and B/C requirements.
- Technology Architecture components of the Architecture Roadmap.
- Refined Phase A deliverables and domain principles, where applicable.

ARTIFACTS / WORK PRODUCTS

- Technology Standards and Technology Portfolio catalogs; Application/Technology matrix; Environments and Locations, Platform Decomposition, Processing, Networked Computing/Hardware, Network and Communications diagrams.

REMEMBER

Define the platform that supports applications and data. Assess services, standards, products and interoperability against stakeholder concerns.

Opportunities & Solutions

OBJECTIVES

Create the initial complete Architecture Roadmap; identify Transition Architectures for incremental value if needed; define overall Solution Building Blocks (SBBs) based on ABBs.

APPROACH

Consolidate cross-domain gaps and dependencies into work packages. Balance transformation readiness, risk and incremental value. Define architecturally significant intermediate states when needed.

INPUTS

- B/C/D draft ADD, requirements, gap analyses and candidate roadmap components.
- Architecture Vision; Statement of Architecture Work; Capability Assessment.
- Governance/management frameworks, planning methods and organizational constraints.
- Change Requests for existing programs/projects; product information and Repository resources.
- Request for Architecture Work; Communications Plan; Organizational Model for EA; Tailored Architecture Framework.

OUTPUTS / DELIVERABLES

- Architecture Roadmap: work packages, dependencies, value, transitions and implementation recommendations (including SBBs).
- Draft ADD: updated domain architectures, stakeholder views and Transition Architectures as needed.
- Draft Architecture Requirements Specification: consolidated gaps, solutions and dependencies.
- Business and IT Capability Assessments.
- Draft Implementation and Migration Plan, including migration strategy.
- Refined Architecture Vision and Statement of Architecture Work, where applicable.

MAIN STEPS

- 1 Determine/confirm key corporate change attributes.
- 2 Determine business constraints for implementation.
- 3 Review and consolidate gap analysis results from Phases B to D.
- 4 Review consolidated requirements across related business functions.
- 5 Consolidate and reconcile interoperability requirements.
- 6 Refine and validate dependencies.
- 7 Confirm readiness and risk for business transformation.
- 8 Formulate Implementation and Migration Strategy.
- 9 Identify and group major work packages.
- 10 Identify Transition Architectures.
- 11 Create the Architecture Roadmap and Implementation and Migration Plan.

ARTIFACTS / WORK PRODUCTS

- Project Context and Benefits diagrams; Implementation Factor catalog; Consolidated Gaps, Solutions, and Dependencies matrix; Architecture Definition Increments Table.

REMEMBER

Initial implementation planning. The Roadmap connects work packages and transitions to the target; the migration plan describes the activity to realize it.

Migration Planning

OBJECTIVES

Finalize the Implementation and Migration Plan and integrate it into the change portfolio. Ensure stakeholders understand the value and cost of work packages and transitions.

APPROACH

Coordinate business planning, EA, portfolio/project management and operations. Prioritize by value, cost, risk and resources; use Transition Architectures as milestones and hand over governance.

INPUTS

- Draft Architecture Roadmap: work packages, transitions and Implementation Factor catalog.
- Draft Implementation and Migration Plan and high-level migration strategy.
- Draft ADD and Requirements Specification; Capability Assessments.
- Business, EA, portfolio/project and operations management frameworks; existing Change Requests.
- Request for Architecture Work; Communications Plan; Organizational Model for EA; Tailored Architecture Framework; Statement of Architecture Work; Architecture Vision; Repository.

MAIN STEPS

- 1 Confirm management framework interactions for the Implementation and Migration Plan.
- 2 Assign a business value to each work package.
- 3 Estimate resource requirements, project timings, and availability/delivery vehicle.
- 4 Prioritize migration projects through cost/benefit and risk assessments.
- 5 Confirm Architecture Roadmap and update Architecture Definition Document.
- 6 Complete the Implementation and Migration Plan.
- 7 Complete the architecture development cycle and document lessons learned.

OUTPUTS / DELIVERABLES

- Approved Implementation and Migration Plan: strategy, project/portfolio breakdown, milestones, resources and costs.
- Finalized Architecture Definition Document, including Transition Architectures as needed.
- Finalized Architecture Requirements Specification.
- Finalized Architecture Roadmap; reusable ABBs.
- Requests for Architecture Work for further cycles, if any.
- Implementation Governance Model, if needed.
- EA Capability Change Requests from lessons learned.

ARTIFACTS / WORK PRODUCTS

- Transition Architecture State Evolution Table; Business Value and Risk Matrix. Reuse and refine the roadmap, work packages and implementation factors developed in E.

REMEMBER

Plan collaboratively with those who fund, build and operate. Governance transitions from architecture development to realization; a governance model is conditional.

Implementation Governance

OBJECTIVES

Ensure implementation projects conform to the Target Architecture. Perform Architecture Governance for solutions and implementation-driven Change Requests.

APPROACH

Implementation runs alongside architecture governance. Use contracts, compliance reviews and change control; guide deployment and operational readiness, then review the implemented solution.

INPUTS

- ADD and Architecture Requirements Specification; Architecture Roadmap.
- Implementation and Migration Plan; Implementation Governance Model.
- Architecture Governance Framework; standard Architecture Contract.
- Statement of Architecture Work; Vision; Capability Assessment; Repository and relevant work requests.
- Organizational Model for EA; Tailored Architecture Framework; external architecture guidance.

MAIN STEPS

- 1 Confirm scope and priorities for deployment with development management.
- 2 Identify deployment resources and skills.
- 3 Guide development of solutions deployment.
- 4 Perform Enterprise Architecture Compliance reviews.
- 5 Implement business and IT operations.
- 6 Perform post-implementation review and close the implementation.

OUTPUTS / DELIVERABLES

- Signed Architecture Contract; Compliance Assessments; Change Requests.
- Deployed architecture-compliant solutions (built by the development process).
- Repository updates, compliance recommendations and dispensations.
- Service requirements, performance recommendations and SLAs.
- Post-implementation updates to Vision and ADD; business/IT operating models and ABBs.

ARTIFACTS / WORK PRODUCTS

- Governance work products: signed Architecture Contract, Compliance Assessments, Change Requests and review decisions. These are deliverables/records, not a mandatory new catalog or diagram set.

REMEMBER

Actual development runs in parallel in the organization's development process. Architectural oversight, contracts and compliance reviews guide conformance.

Architecture Change Management

OBJECTIVES

Maintain the architecture development cycle, execute the Architecture Governance Framework and ensure the EA Capability meets current requirements.

APPROACH

Monitor value, performance, risks and changing business/technology needs. Assess change through governance: maintenance may update the architecture; major change can trigger a new ADM cycle.

INPUTS

- Current ADD, Requirements Specification and Architecture Roadmap.
- Change Requests from technology change, business change and lessons learned.
- Signed Architecture Contract; Compliance Assessments.
- Implementation Governance Model and Implementation and Migration Plan.
- Architecture Vision, Statement of Architecture Work, framework and Repository.
- Request for Architecture Work; Organizational Model for EA; external architecture guidance.

MAIN STEPS

- 1 Establish value realization process.
- 2 Deploy monitoring tools.
- 3 Manage risks.
- 4 Provide analysis for architecture change management.
- 5 Develop change requirements to meet performance targets.
- 6 Manage governance process.
- 7 Activate the process to implement change.

OUTPUTS / DELIVERABLES

- Architecture updates for maintenance changes.
- Framework and principle changes for maintenance changes.
- New Request for Architecture Work for major changes and a further cycle.
- Statement of Architecture Work, Architecture Contract and Compliance Assessments, updated if needed.

ARTIFACTS / WORK PRODUCTS

- Change and monitoring records: value/performance assessments, Change Requests and governance decisions. Update affected models and deliverables; no fixed new diagram set is required.

REMEMBER

Monitor realized value and changing needs. The governance process decides how changes are handled; maintenance and major changes have different outputs.

Architecture Requirements Management

OBJECTIVES

Sustain requirements management across relevant ADM phases. Manage identified requirements and make the applicable requirements available to each phase.

APPROACH

Manage the flow and traceability of requirements, including changes and impacts. Relevant ADM phases decide priorities and implementation; Requirements Management records and coordinates them.

INPUTS

- Populated Architecture Repository; Organizational Model for EA; tailored framework.
- Statement of Architecture Work; Architecture Vision.
- Architecture requirements in the Architecture Requirements Specification.
- Requirements Impact Assessment.

OUTPUTS / DELIVERABLES

- Requirements Impact Assessment.
- Updated Architecture Requirements Specification, if needed.
- Architecture Requirements Repository updated during the process (step 8).

MAIN STEPS

- 1 ADM phase: identify and document requirements.
- 2 Requirements Management: establish the baseline and record agreed priorities.
- 3 Requirements Management: monitor baseline requirements.
- 4 ADM phase: identify new/changed requirements; remove, add or modify and reassess priorities.
- 5 Requirements Management: record changes/priorities, manage conflicts through phases and generate an impact assessment.
- 6 ADM phase: assess current/prior-phase impacts; decide change or deferral; issue impact assessment version n+1.
- 7 ADM phase: implement requirements arising from Phase H.
- 8 Requirements Management: update the Requirements Repository, including affected stakeholder views.
- 9 ADM phase: implement change in the current phase.
- 10 ADM phase: assess/revise prior gap analyses and address gap requirements.

ARTIFACTS / WORK PRODUCTS

- Requirements catalog: requirements, assumptions, constraints and gaps. Requirements Impact Assessment records affected phases, priorities and consequences; trace changes to stakeholder views.

REMEMBER

Requirements Management owns steps 2, 3, 5 and 8; the ADM phases own the others. The repository is updated during the process; the formal outputs are the impact assessment and updated specification.